I Guess We'd Better Talk About PDF Accessibility

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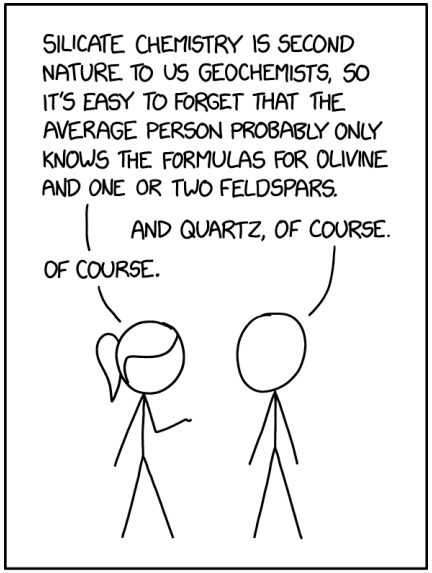
IMACC Conference
April 2026

Speaker Background

- I did elementry school homework in Word 1.0 on a Macintosh SE
- I made my first web page in 1996
  - We thought <blink> was pretty cool
  - Alt text used to be for people with slow internet connections
- I was in Harper's pilot group for Blackboard Ally in Fall 2020



EVEN WHEN THEY'RE TRYING TO COMPENSATE FOR IT, EXPERTS IN ANYTHING WILDLY OVERESTIMATE THE AVERAGE PERSON'S FAMILIARITY WITH THEIR FIELD.

https://xkcd.com/2501/

Universal Design for Accessibility

- Accessibility is Good
- Undiagnosed or Undisclosed Disabilities
  - Disclosing Disabilities is a Pain, Actually
- Curb-Cut Effect

🐘 🐘 🐘 BUT 🐘 🐘 🐘

- Competing Access Needs

Accessible Documents

- Three Goals
  - An overall accessible course
  - Accessible documents
  - Documents that software tools **think** are accessible
- These goals can be at odds!

What’s an Accessible Document?

- Screen reader friendly structure
  - Text is actually text
  - Images have “alt text” (to be read when the image would be shown)
  - Tables have headers (to help give them meaning)
- Colorblind friendly
  - Colorblindness is *really common* (esp. in people with one X chromosome) and may not be registered as a “disability.”
  - Don’t encode important information ONLY with color
    - “Red dashed graph vs green solid graph” not “red graph vs green graph”
  - High contrast

PDF Considered Harmful?

- PDF is not a Web format
  - *Web pages* should be XML/HTML
- Documents designed for print should be supplied as PDF
  - Consistent presentation
  - Platform independence
  - Layout conveys information
- Documents intended for on-screen viewing should not
  - More on this later
- PDF is a proprietary format 😤

Speaking of proprietary formats...

You need Adobe Acrobat Pro

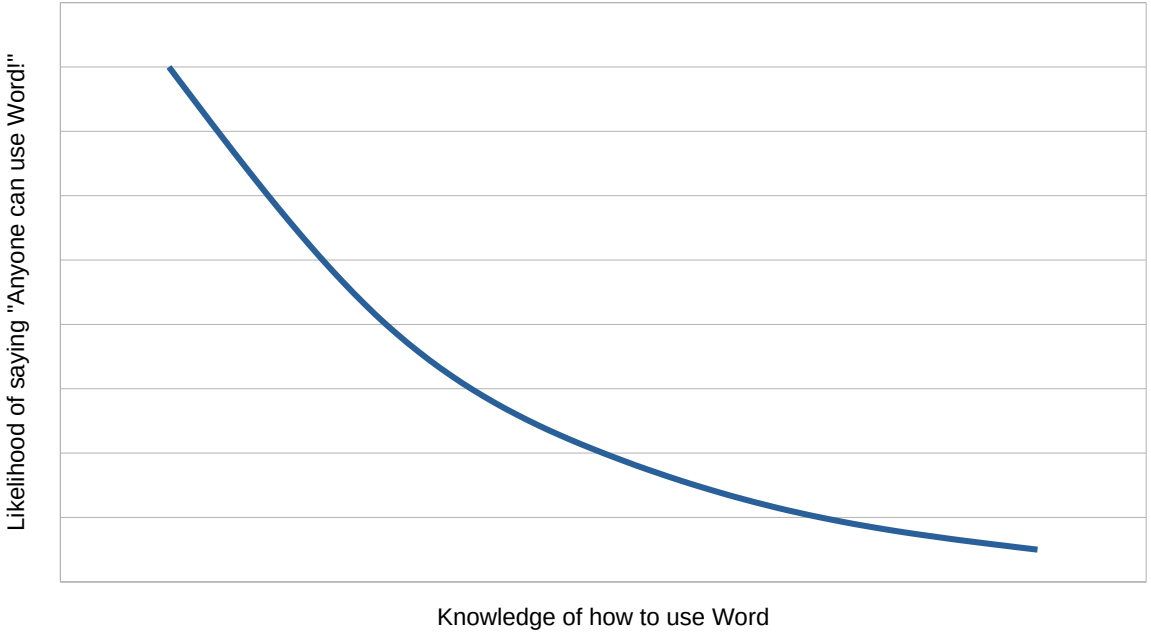
- The last version of Acrobat with a perpetual license was Acrobat Pro 2020. Sales ended in 2024.
- Do with this information what you will.

Why Do You Need Acrobat?

- Fix scanned documents
- Fix annotated documents
- Fix old PDF files
- **Figure out what the problem actually is**

The Good News

- Microsoft Word makes this easy
  - If you actually learn how to use Word



How to Use Word

- “Outline” view
  - Your document should make sense from the headers.
- “Table Design” ribbon
  - “Header,” “First Column,” or both
  - Tables are for tables – consider using columns or tabs for layout
- Images
  - In-Line
  - Alt Text
- Check Accessibility
  - Happens automatically when you try to save to PDF!
- “Title” is carefully hidden

Alt Text for Images

Who will “see” (hear) your alt text?

- Lots of people use a screen reader!
- Reading-Based Disabilities (Dyslexia)
- Attention-Based Disabilities (ADHD)
- Multi-Taskers
- Vision-Based Disabilities
  - Visually Impaired
  - Blind (Low Vision)
  - Blind (Sightless)

Alt Text for Images – Why Is This Image Here?

- What information should a sighted person get by looking at the image?
- Common kinds of information in math images:
  - Numbers that can be read from the graph
    - “A line with slope -1/2 and y-intercept (0,4).”
  - General shape of the graph
    - “A parabola and a line. They intersect at two points, marked as P and Q.”
  - Data also shown elsewhere
    - “A histogram for the preceding data table. It is approximately bell-shaped.”
- Think about what you want to **communicate**.

But what if —

- What if the information is, “This is what this picture looks like!”
  - “Image Described In Text”
  - “A graph of the preceding equation”
  - “Diagram for [purpose]”
- Don’t be afraid to mark it decorative
  - If you have a blind (sightless) student, you will work with disability services to develop tactile alternatives.
  - The TI-84 is a useless device to anyone who can’t see your TI-84 screen shots. Mark them decorative.
    - DESMOS is the accessible alternative for blind (sightless) users

Don’t Panic!

- **Bad alt text is better than no alt text.**
- Short alt text is better than excessively long alt text.
- You are **probably** already describing the image in text.
- You can edit later.

How to Make Your Scanned Notes Accessible

Posting Scanned Notes

- Posting your class notes makes your **course** more accessible!
- A hand-written document will never be an accessible **document**.
- The **information** should be available in your course in an accessible format.
  - For an in-person class, this is probably your actual **lecture**!
- But, you *can* get your accessibility checker off your back.

Making Course Notes Accessible

- Accessibility checker may not like “Scanned and OCR’d” notes
- It likes “annotated” (on a computer) notes better
- If you are annotating a PDF, the PDF you start with should be accessible.
- This is where you need Acrobat!
- Balance time vs. quality

LaTeX Accessibility

The Good News

- Accessible LaTeX output now exists!
- (As of LaTeX 3.0; make sure you're up to date.)
- … Kind of.

The Bad News

- As of right now, Tagged PDF is incompatible with basically all packages.
  - That’s an exaggeration (but not much)
  - TikZ works!
  - If it messes with tables, headings, or anything numbered, it’s probably broken
- Most of them quietly break the tagging.
- Some of them just won’t run.

The Ugly News—Let’s Talk PDF Standards

PDF 1.7 – PDF UA/1

- 2008 standard
- Supported by most accessibility checkers
- Expects alt tags for math formulas
- Weakly supported by LaTeX

PDF 2.0 – PDF UA/2

- 2020 standard
- Supported by few accessibility checkers
- Embeds MathML for math formulas
- Well supported by LaTeX
- As of 4/8, Blackboard claims “6 month timeline” to support

Getting Started: PDF 1.7 – PDF/UA-1

- Make sure your LaTeX installation is up to date
- Add code at the *beginning* of your TeX document.

```
\DocumentMetadata{
  lang        = en,
  Pdfversion  = 1.7,
  pdfstandard = ua-1,
  tagging=on,
  tagging-setup={math/mathml/luamml/load=false}
}
```

- Run with LuaLaTeX

Getting Started: PDF 2.0 – PDF/UA-2

- Make sure your LaTeX installation is up to date
- Add code at the *beginning* of your TeX document.

```
\DocumentMetadata{
  lang        = en,
  pdfstandard = ua-2,
  tagging=on,
  tagging-setup={math/setup=mathml-SE}
}
```

Run with LuaLaTeX

Checking Accessibility

- For PDF 1.7 – PDF/UA-1, use your LMS
- For PDF 2.0 – PDF/UA-2, use VeraPDF (linked)
- For either – open in Acrobat to examine tagging structure

Alt Tags for Images

- The following commands support alt tagging:
  - `\includegraphics`
  - `\tikzpicture` (or the environment version)
- Include in the optional argument
- Alt text: `alt={alt text}`
- Decorative image: `artifact`
- Image of text: `actualtext=text`

Fun With Tables

- Accessible tables need headers.
  - `\tagpdfsetup{table/header-rows={1}}`
- Or, if they’re not *really* tables, they should be marked as display.
  - `\tagpdfsetup{table/tagging=presentation}`
- In the preamble (applied to all tables) or before each table

Do *You* Have 150 Worksheets?

- Use the command line! Run as a batch!
- Windows Powershell example (with TeX installation in $Path)

```
$files = Get-ChildItem -Path "." -Filter "In-Class*.tex"
foreach ($file in $files) {
    & lualatex.exe $($file.Name)
}
```

Going Forward

- LaTeX is a typesetting program.
  - Typesetting means **PRINT**
  - If it’s not basically a print document, it shouldn’t be in LaTeX
- Use PreTeXt for e-books
  - HTML is accessible
  - You can also get ebook formats (sweet!)
  - If you use TikZ, you kinda need to be on Linux

Oh No, My OER!

Don’t Panic!

- It might not be as bad as you think.
- Run it through Acrobat’s accessibility checker and auto-fix.
  - Be careful! Acrobat occasionally eats things
- If you have Word files, use the Word tricks.
- If you’ve *adopted* OER, check for an updated version.
- Done is good – aim for improvement, not perfection
- Collaboration! You’re not alone.

Online Homework for OER?

- MyOpenMath is good on accessibility.
  - Alt text your images.
  - Don’t commit evil in your code.
  - Write accessible HTML.
- I made OER homework in my LMS!
  - Why? No, really, why?
  - HTML best practices will still help you, though.
- Alt text is easier here
  - Give enough information to solve the problem!

Links and Code Samples

- Get my links and sample files on my github

https://github.com/amasaros/pdftalk2026/

That’s All, Folks!

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Github: https://github.com/amasaros/pdftalk2026/

File Outlining Home Insert Draw Design Layout References Mailings Review View

Read Mode | Print Layout | Web Layout | Outline | Draft — Views

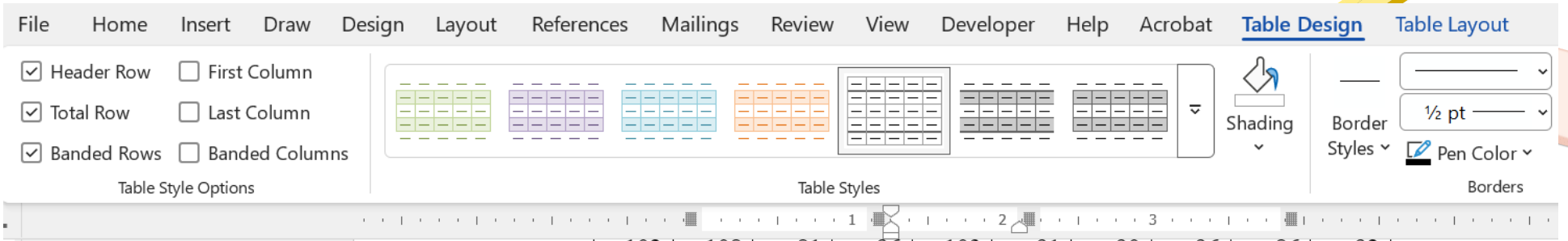
Focus | Immersive Reader — Immersive

Ruler | Gridlines | Navigation Pane — Show

Zoom | 100% | One Page | Multiple Page | Page Width — Zoom

Section 3.3: Formal Logic

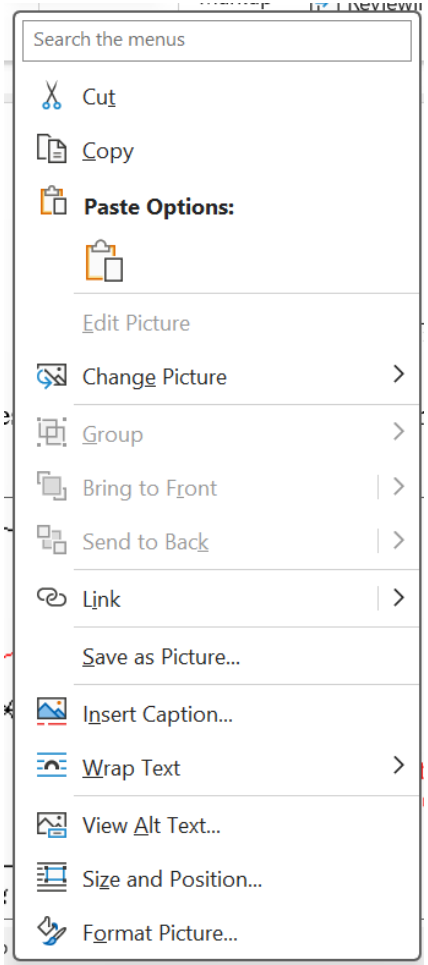
- Concept 1 – Truth Values
- Concept 2 – Simple and Compound Statements
- Concept 3 – Symbolizing Statements
- Concept 4 – Truth Tables
- Concept 5 – Negation of a Statement
- Concept 6 – Conjunction
- Concept 7 – Disjunction
- Concept 8 – Conditional and Biconditional
- Concept 9 – Truth Tables to Symbolize Statements



102	108	81	96	103	91	90	96	86	92
84	92	90	103	115	93	85	116	87	106
85	88	106	104	102	98	116	107	102	89

Frequency and Relative Frequency for IQ Scores

IQ	Frequency	Relative Frequency
63.5 – 70.5	2	0.04 or 4%
70.5 – 77.5	2	0.04 or 4%
77.5 – 84.5	5	0.10 or 10%
84.5 – 91.5	9	0.18 or 18%
91.5 – 98.5	11	0.22 or 22%
98.5 – 105.5	9	0.18 or 18%
105.5 – 112.5	6	0.12 or 12%
112.5 – 119.5	4	0.08 or 8%
119.5 – 126.5	1	0.02 or 2%
126.5 – 133.5	1	0.02 or 2%
Total	**50**	**1.00 or 100%**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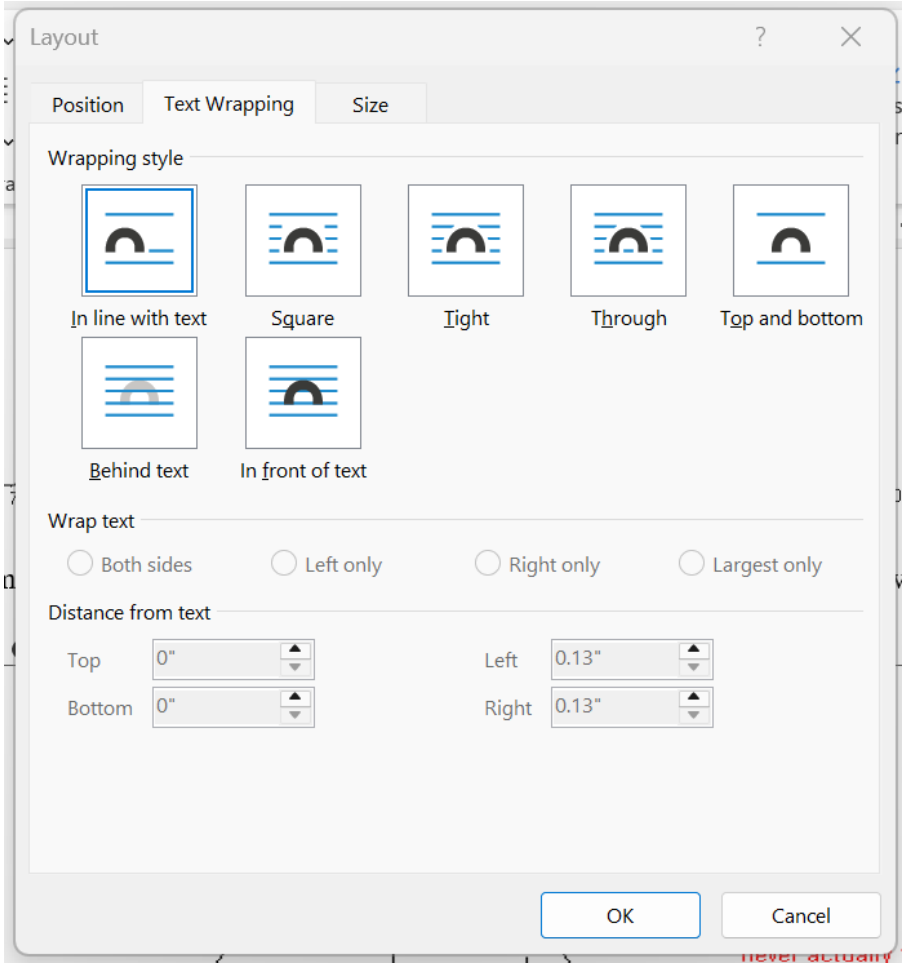
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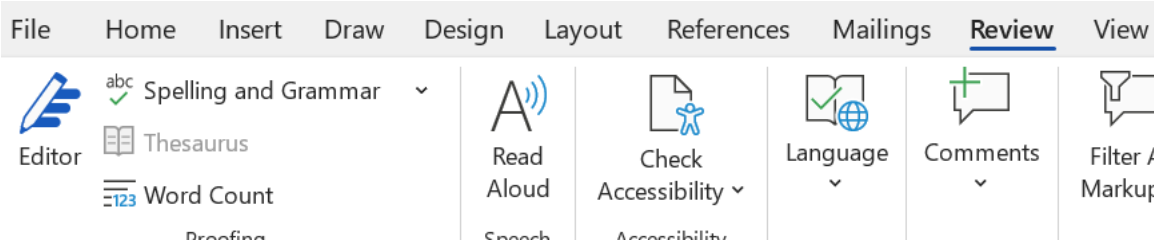
OK
Cancel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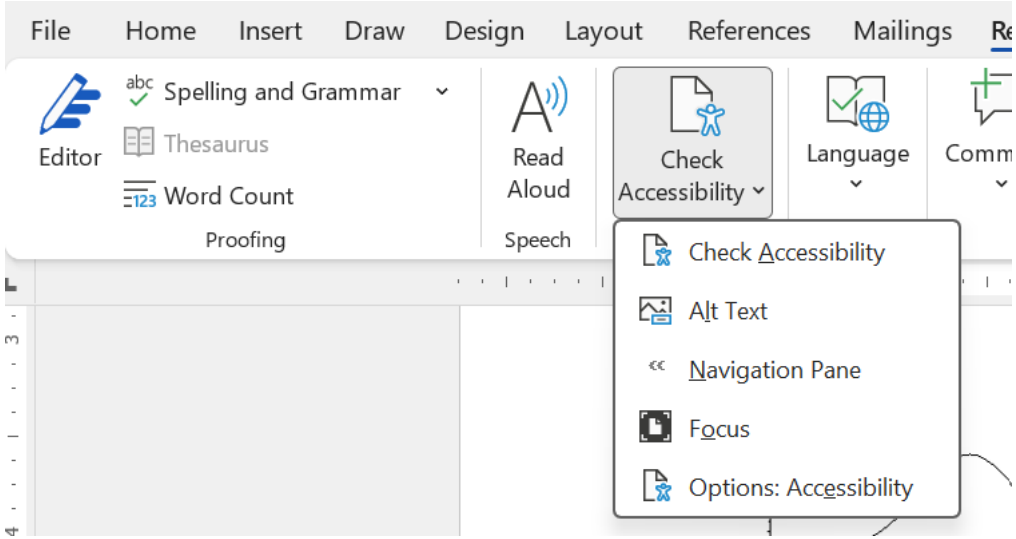
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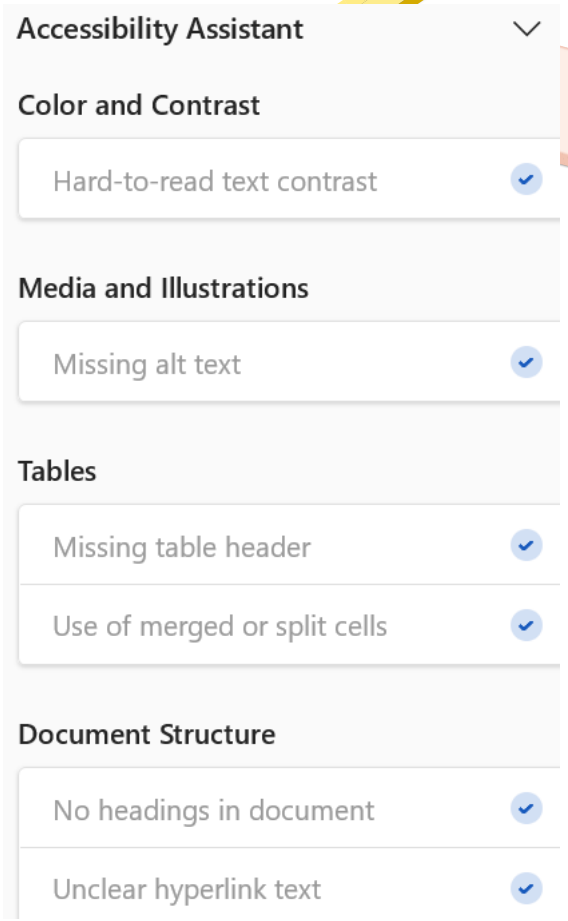
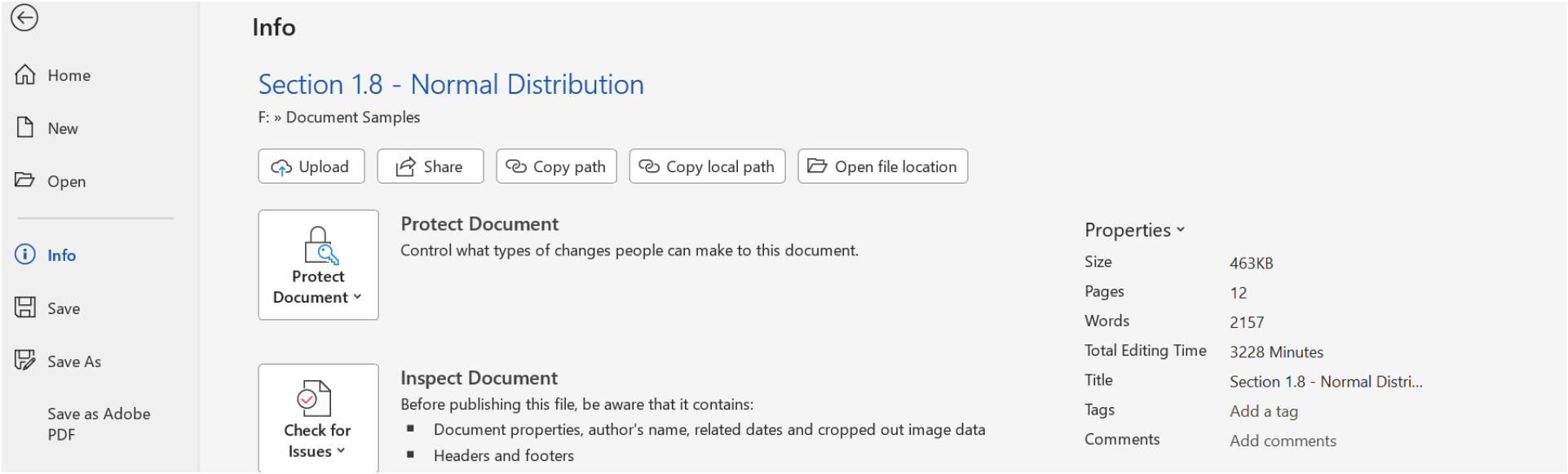

Accessibility Assistant
Color and Contrast
Hard-to-read text contrast
Media and Illustrations
Missing alt text
Tables
Missing table header
Use of merged or split cells
Document Structure
No headings in document
Unclear hyperlink text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Protect Document
Control what types of changes people can make to this document.
Check for Issues
Inspect Document
Before publishing this file, be aware that it contains:
Document properties, author's name, related dates and cropped out image data
Headers and footers
Properties
Size 463KB
Pages 12
Words 2157
Total Editing Time 3228 Minutes
Title Section 1.8 - Normal Distri...
Tags Add a tag
Comments Add comments